

A STUDENTS' INDUSTRIAL WORK EXPERIENCE SCHEME

(SIWES) TECHNICAL REPORT

ON

DATA SCIENCE

CARRIED OUT AT

EDDY-BLACK TECHNOLOGY

IBIAM STREET, UYO, AKWA IBOM STATE

BY

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**IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE
STUDENTS' INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES) OF
THE INDUSTRIAL TRAINING FUND (ITF)**

October, 2026

CERTIFICATION

This is to certify that this report on the Students' Industrial Work Experience Scheme (SIWES) undertaken at Eddy-Black Technology, Ibiam Street, Uyo, Akwa Ibom State, was carried out by **ESSIEN EMMANUEL ITA** with Registration Number **23/SC/CO/031**, and has been read and approved as meeting the requirements of the Department and the Industrial Training Fund (ITF) for the award of the relevant certificate.

.....

Mr Asuquo, Idongesit Isaac

Industry-Based Supervisor

Date:

.....

Institution-Based Supervisor

Date:

.....

SIWES Coordinator

Date:

DEDICATION

This report is dedicated to Almighty God for His grace, wisdom, and sustenance throughout the period of my Students' Industrial Work Experience Scheme (SIWES). It is also dedicated to my parents and family members, whose unwavering support, prayers, and encouragement have been the foundation of my academic and professional pursuits, and to every student striving to bridge the gap between theoretical knowledge and practical industry skills in the field of Data Science.

ACKNOWLEDGEMENT

I give all glory and honor to God Almighty for granting me the strength, knowledge, and good health to undergo and complete my Students' Industrial Work Experience Scheme (SIWES) training successfully.

My profound gratitude goes to my industry-based supervisor, **Mr Asuquo, Idongesit Isaac**, for his patient guidance, technical mentorship, and constructive corrections throughout my

attachment at Eddy-Black Technology. His willingness to share practical knowledge in Data Science greatly shaped the quality of the work presented in this report.

I also wish to specially appreciate the Head of Technical Team, **Miss Marvelous Emmanuel John**, for her exceptional leadership, hands-on training sessions, and the enabling environment she created that allowed me to explore real-world data science tools and projects with confidence.

My sincere appreciation goes to the entire management and staff of Eddy-Black Technology, Ibiam Street, Uyo, for accepting me as an intern and providing me with the resources, tools, and mentorship necessary for a fulfilling industrial training experience.

I equally acknowledge my institution-based supervisor and the SIWES coordinating unit of my department for their supervision, monitoring, and useful academic guidance during this attachment. Special thanks also go to the Industrial Training Fund (ITF) for establishing and sustaining this all-important scheme that bridges the gap between classroom theory and industrial practice.

Finally, I am deeply grateful to my parents, siblings, and friends for their moral, financial, and spiritual support throughout the duration of this programme. May God bless you all abundantly.

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ABSTRACT

This report documents the practical experience gathered during the Students' Industrial Work Experience Scheme (SIWES) undertaken by Essien Emmanuel Ita, with Registration Number 23/SC/CO/031, at Eddy-Black Technology, Ibiam Street, Uyo, Akwa Ibom State,

under the Data Science unit of the organisation. The training exposed the student to the fundamental and applied concepts of data science, including data collection, data cleaning, exploratory data analysis, feature engineering, machine learning model building, and model evaluation. As the core deliverable of the attachment, a real, end-to-end machine learning project was designed and implemented: a Customer Churn Prediction Model intended to help telecommunication-type service businesses identify customers who are likely to discontinue a service or subscription. The project made use of Python programming language and its data science ecosystem, comprising Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn, to build, train and compare two classification models — Logistic Regression and Random Forest — on a structured customer dataset containing features such as tenure, monthly charges, contract type, internet service type, technical support subscription, and payment method. The models were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and Receiver Operating Characteristic (ROC) curves. Results showed that the Logistic Regression model attained an accuracy of approximately 82.3%, while the Random Forest model attained approximately 80.3%, with month-to-month contracts, higher monthly charges, and shorter tenure emerging as the strongest predictors of churn. The report further presents the organisational profile of Eddy-Black Technology, the methodology and system architecture adopted for the project, the weekly log of activities undertaken, challenges encountered during the training, and recommendations for the improvement of future SIWES placements in data science-related establishments. It is concluded that the SIWES programme significantly enhanced the student's practical competence in data science and machine learning, complementing the theoretical foundation acquired in the classroom.

CHAPTER ONE

INTRODUCTION

1.1 Background to SIWES

The Students' Industrial Work Experience Scheme (SIWES) is a skills training programme established by the Federal Government of Nigeria in 1973 and managed by the Industrial Training Fund (ITF) in collaboration with the National Universities Commission (NUC), the National Board for Technical Education (NBTE), and other regulatory bodies (Industrial Training Fund [ITF], 2019). The scheme was designed to expose students in higher institutions, particularly those in science, technology, engineering, and management-related disciplines, to the practical industrial environment they are expected to encounter after graduation.

The rationale behind SIWES is anchored on the observation that theoretical knowledge acquired in classrooms is often insufficient without complementary hands-on, practical exposure to real work situations. SIWES therefore bridges the gap between theory and practice by attaching students to relevant industries, government establishments, and organisations for a specified period, typically ranging from four to six months, depending on the institution and course of study.

For a Computer Science/Data Science-oriented student such as the author of this report, SIWES provides an invaluable opportunity to interact with real datasets, business problems, and industry-standard tools that are often difficult to fully simulate in a classroom setting. This report presents the training undertaken and the project executed by **Essien Emmanuel Ita 23/SC/CO/031** during his industrial attachment at Eddy-Black Technology, Ibiam Street, Uyo, Akwa Ibom State, under the supervision of Mr Asuquo, Idongesit Isaac, and the technical mentorship of Miss Marvelous Emmanuel John, Head of the Technical Team.

1.2 Aims and Objectives of SIWES

According to the Industrial Training Fund (2019), the broad objectives of the Students' Industrial Work Experience Scheme include the following:

- To provide students with the opportunity to apply theoretical knowledge acquired in the classroom to real industrial work situations, thereby bridging the gap between academic training and practical application.

- To expose students to work methods, tools, technologies, and techniques in handling equipment and machinery that may not be readily available in their institutions.

- To prepare students for the industrial work environment they are likely to meet after graduation, thereby enhancing their employability and reducing the pressure and cost of retraining fresh graduates.
- To provide an avenue for students to develop occupational competencies, professional ethics, and interpersonal skills required for effective functioning in the world of work.
- To enlist and strengthen employers' involvement in the entire educational process, thereby preparing students for the labour market.

In the specific context of this report, the training was designed to expose the student to practical Data Science workflows, including data acquisition, cleaning, exploratory analysis, statistical reasoning, machine learning model development, and the communication of analytical results to non-technical stakeholders — all of which are documented extensively in subsequent chapters.

1.3 Profile of Eddy-Black Technology

Eddy-Black Technology is a technology-driven establishment located at Ibiam Street, Uyo, Akwa Ibom State, Nigeria. The organisation renders a range of information technology services, including software development, data analytics and data science consultancy, systems administration, digital training, and technology-based solutions to small and medium-sized businesses within and outside Akwa Ibom State.

The organisation runs a structured technical unit that handles client-facing data projects, ranging from business intelligence dashboards to predictive analytics solutions, and also operates an internship/training arm through which students undergoing SIWES and other young technology enthusiasts are mentored in in-demand digital skills such as data science, web development, and cloud computing. The company's approach emphasises hands-on, project-based learning, which aligns closely with the practical objectives of the SIWES programme.

1.4 Organisational Structure of the Establishment

Eddy-Black Technology operates a fairly lean but effective organisational structure suited to a growing technology company. At the apex of the organisation is the

Management/Executive arm, which oversees overall strategy and client relationships. Beneath this is the Technical Department, headed by the Head of Technical Team, Miss Marvelous Emmanuel John, who coordinates all technical units including the Data Science and Analytics unit, the Software Development unit, and the IT Support/Infrastructure unit.

Industrial trainees and interns admitted into the organisation, including the author of this report, are typically attached to a specific unit based on their course of study and area of interest, and are supervised directly by an assigned mentor — in this case, Mr Asuquo, Idongesit Isaac — who reports progress to the Head of Technical Team. This structure ensured that the student received close mentorship, structured tasks, and constructive feedback throughout the duration of the attachment.

1.5 Aim and Objectives of the Report

This report aims to document, in a structured and academically acceptable manner, the practical experience, skills acquired, and project executed by the student during the SIWES training at Eddy-Black Technology. Specifically, the report seeks to:

- Document the organisational profile and structure of Eddy-Black Technology as the host establishment.
- Review relevant literature on Data Science, its processes, tools, and applications.
- Describe the methodology and training activities undertaken during the SIWES period.
- Present a real, practical Data Science project — a Customer Churn Prediction Model — undertaken as part of the training, complete with diagrams, code, and results.
- Highlight the challenges encountered during the training and proffer recommendations for the improvement of future SIWES placements.

1.6 Scope of the Report

The scope of this report is limited to the activities, training, and project work carried out by the student during the SIWES attachment period at Eddy-Black Technology. While broader concepts of data science and machine learning are reviewed for context, the practical project presented in Chapter Four is scoped specifically to a customer churn prediction problem using a structured, tabular dataset, and does not extend to unstructured data domains such as

image or natural language processing, which, though relevant to data science generally, fell outside the immediate scope of the training and project timeline.

LITERATURE REVIEW

2.1 Concept and Definition of Data Science

Data Science is a multidisciplinary field that combines statistics, computer science, and domain expertise to extract meaningful insights and knowledge from structured and unstructured data (Provost & Fawcett, 2013). It encompasses the entire process of collecting, cleaning, exploring, modelling, and communicating data-driven insights to support decision

making. Han, Kamber and Pei (2011) describe data mining, a core component of data science, as the process of discovering interesting patterns and knowledge from large amounts of data, drawing from areas such as database systems, statistics, and machine learning.

Kotu and Deshpande (2019) further characterise data science as an applied discipline, emphasising that its value is realised not merely through sophisticated algorithms, but through the practical application of these algorithms to solve genuine business and societal problems. This practical orientation of data science is precisely what SIWES seeks to expose students to — the translation of theoretical models into deployable, business-relevant solutions.

2.2 The Data Science Process

Several frameworks have been proposed to formalise the data science workflow, the most widely adopted of which is the Cross Industry Standard Process for Data Mining (CRISP DM), developed by Chapman et al. (2000). The CRISP-DM framework decomposes a typical data science project into six iterative phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment.

This framework, adapted slightly for the purpose of this report into an eight-stage life cycle (business understanding, data collection, data cleaning and pre-processing, exploratory data analysis, feature engineering, model building, model evaluation, and deployment/reporting), was the guiding methodology adopted for the project executed during the SIWES training, as elaborated in Chapter Three of this report.

2.3 Tools and Technologies used in Data Science

A wide variety of tools and programming languages are employed in modern data science practice. Python has emerged as one of the most dominant languages in the field owing to its simplicity, readability, and the richness of its data science ecosystem (VanderPlas, 2016). Central to this ecosystem are libraries such as Pandas, which provides high-performance, easy-to-use data structures for data manipulation and analysis (McKinney, 2010); NumPy, which supports efficient numerical computation; and Matplotlib and Seaborn, which provide extensive data visualisation capabilities.

For machine learning specifically, Scikit-learn has become a de facto standard library within the Python ecosystem, offering simple and efficient tools for predictive data analysis, including classification, regression, clustering, and model evaluation utilities (Pedregosa et al., 2011). These tools formed the core technology stack used during the SIWES training and the project documented in this report.

2.4 Overview of Machine Learning

Machine learning is a subfield of artificial intelligence concerned with the development of algorithms that can learn patterns from data and make predictions or decisions without being explicitly programmed for every scenario (Géron, 2019). Machine learning algorithms are broadly categorised into supervised learning, unsupervised learning, and reinforcement learning. Supervised learning, which was the category of learning applied in the project presented in this report, involves training a model on a labelled dataset — that is, a dataset in which the outcome or target variable is already known — so that the model can learn the relationship between input features and the target.

Two supervised learning algorithms were employed in the project documented in Chapter Four: Logistic Regression and Random Forest. Logistic Regression is a statistical model that estimates the probability of a binary outcome using a logistic (sigmoid) function of a linear combination of input features, and remains one of the most interpretable and widely used classification algorithms in industry (Hosmer, Lemeshow, & Sturdivant, 2013).

Random Forest, introduced by Breiman (2001), is an ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of the classes (for classification tasks) predicted by the individual trees, generally yielding improved predictive accuracy and control over overfitting compared to a single decision tree.

2.5 Application of Data Science to Customer Churn Prediction Customer churn

— the phenomenon whereby customers discontinue their relationship with a service provider — is one of the most extensively studied application areas of data science and machine learning in industry, particularly within the telecommunications, banking, and subscription-service sectors. Verbeke, Martens, Mues and Baesens (2011) note that even a small reduction in customer churn can translate into substantial financial gains for a business, because retaining an existing customer is typically far less expensive than acquiring a new one.

Ahmad, Jafar and Aljoumaa (2019) developed and compared several machine learning models, including Random Forest and Gradient Boosted Machines, for churn prediction in a telecommunications big-data environment, and demonstrated that ensemble tree-based methods generally perform competitively on this class of problem, particularly when combined with well-engineered features such as customer tenure, contract type, and billing information. These findings directly informed the choice of features and models adopted in the customer churn prediction project undertaken during the SIWES training at Eddy-Black Technology, as presented in Chapter Four of this report.

It is important to note that customer churn prediction problems, in practice, typically exhibit class imbalance — that is, the number of customers who churn is usually much smaller than the number who do not — a characteristic that was equally observed in the dataset used for this project and which necessitated careful selection of evaluation metrics beyond simple accuracy, such as precision, recall, and the area under the Receiver Operating Characteristic curve (Verbeke et al., 2011).

CHAPTER THREE

TRAINING UNDERGONE AND METHODOLOGY

3.1 Departments and Units Attached To

During the period of industrial attachment, the student was posted to the Data Science and Analytics unit of the Technical Department at Eddy-Black Technology, under the direct supervision of Mr Asuquo, Idongesit Isaac, and the overall coordination of the Head of Technical Team, Miss Marvelous Emmanuel John. In addition to the core Data Science unit, the student also had brief cross-functional exposure to the Software Development unit, which provided useful context on how data science models are eventually integrated into deployable software applications.

The training was structured around a blend of guided instructional sessions, self-paced online learning (using resources recommended by the supervisor), and hands-on project work, culminating in the design and implementation of a complete, end-to-end machine learning project, which is presented in full in Chapter Four of this report.

3.2 Summary of Weekly Activities

Table 3.1 presents a summary of the major activities undertaken by the student on a weekly basis throughout the sixteen-week period of industrial attachment at Eddy-Black Technology.

Table 3.1: Summary of Weekly Activities Undertaken During SIWES

Week	Activity Description
1	Orientation, introduction to the organisation's policies, tools, and the Data Science unit's workflow.
2	Introduction to Python programming refresher; installation and configuration of Anaconda, Jupyter Notebook, and VS Code.
3	Hands-on training on Pandas and NumPy for data manipulation, indexing, filtering, and aggregation.
4	Data visualisation training using Matplotlib and Seaborn; practice on sample retail and telecom datasets.
5	Introduction to statistics for data science: measures of central tendency, dispersion, correlation, and hypothesis testing.
6	Study of data cleaning techniques: handling missing values, duplicates, outliers, and inconsistent data formats.

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Week	Activity Description
7	Introduction to Machine Learning concepts: supervised vs unsupervised learning, bias-variance trade-off, and model evaluation metrics.
8	Practical sessions on Scikit-learn : building simple regression and classification models on toy datasets.
9	Commencement of the assigned project: problem definition and business understanding for a Customer Churn Prediction system.
10	Data simulation/collection and exploratory data analysis (EDA) for the churn prediction project.
11	Feature engineering, encoding of categorical variables, and feature scaling for model readiness.
12	Model building: training Logistic Regression and Random Forest classifiers on the pre-processed dataset.
13	Model evaluation using confusion matrix, ROC-AUC, precision, recall, and F1-score; hyperparameter tuning.
14	Preparation of visual dashboards/reports summarising project findings for presentation to the Head of Technical Team.
15	Presentation of the completed churn prediction project to the technical team; feedback and refinement of the model.
16	Documentation of the project, compilation of the SIWES log book, and preparation of this technical report.

3.3 Software and Tools Used During Training

The following software, programming languages, and tools were utilised during the course of the industrial training and the execution of the project presented in this report:

- Python 3 — the primary programming language used for data manipulation, analysis, and machine learning model development.
- Jupyter Notebook and Visual Studio Code (VS Code) — integrated development environments used for writing and testing Python code interactively.
- Pandas and NumPy — libraries used for data loading, cleaning, transformation, and numerical computation.
- Matplotlib and Seaborn — libraries used for statistical and exploratory data visualisation.
- Scikit-learn — the primary machine learning library used for feature encoding, model training, and model evaluation.
- Microsoft Excel — used occasionally for quick data inspection and validation of results.

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- Git and GitHub — used for version control and collaborative code management within the technical team. Github repo link:

<https://github.com/emmanuelitaessien02-max/SIWES-Data-Science-Practical>

3.4 Methodology Adopted

The methodology adopted for the execution of the practical project during the SIWES training followed an adapted version of the widely recognised CRISP-DM (Cross Industry Standard Process for Data Mining) framework (Chapman et al., 2000). The eight-stage life cycle illustrated in Figure 3.1 guided every phase of the project, from initial problem understanding through to the final reporting of results.

Figure 3.1: The Data Science Project Life Cycle Adopted During SIWES Training

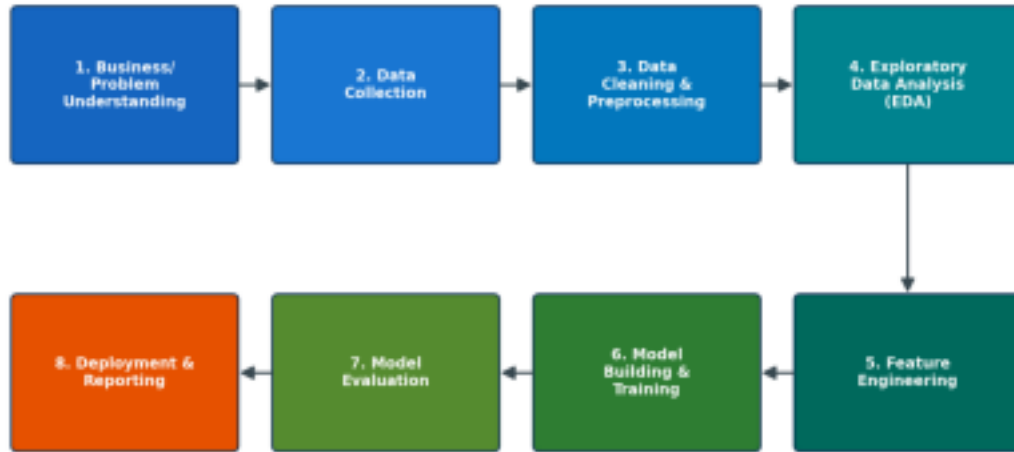


Figure 3.1: The Data Science Project Life Cycle Adopted During SIWES Training

Each stage of this life cycle is briefly described as follows: Business Understanding involved clarifying the business problem — predicting which customers are likely to churn; Data Collection involved obtaining a structured customer dataset with relevant behavioural and billing attributes; Data Cleaning and Pre-processing involved handling inconsistencies and preparing the data for analysis; Exploratory Data Analysis (EDA) involved visually and statistically examining relationships within the data; Feature Engineering involved encoding categorical variables and scaling numerical variables; Model Building and Training involved fitting Logistic Regression and Random Forest classifiers; Model Evaluation involved assessing model performance using appropriate metrics; and Deployment and Reporting involved summarising findings into an actionable business report.

3.5 Proposed System Architecture

Figure 3.2 presents the proposed system architecture designed for the customer churn prediction solution developed during the training. The architecture is organised into distinct layers — a data ingestion and cleaning layer, an exploratory data analysis layer, a feature engineering layer, a model training layer, a model evaluation layer, and finally, an output layer that feeds predictions into a business decision-making process, such as a customer retention team.

Figure 3.2: Proposed System Architecture for the Customer Churn Prediction Solution

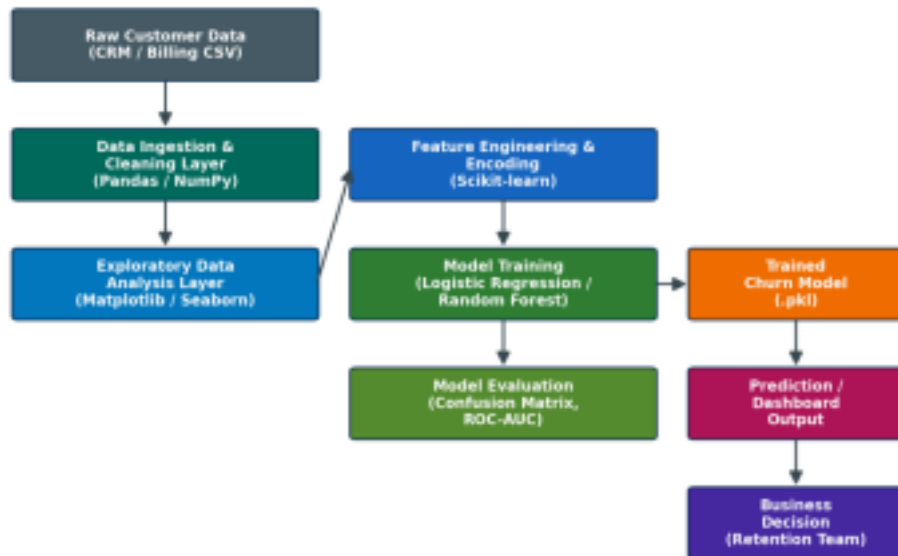


Figure 3.2: Proposed System Architecture for the Customer Churn Prediction Solution

This layered architecture ensures a clear separation of concerns, making the resulting system easier to maintain, extend, and eventually integrate into the organisation's broader software infrastructure, an approach consistent with recommended practice in applied machine learning system design (Géron, 2019).

CHAPTER FOUR

PROJECT WORK: CUSTOMER CHURN PREDICTION MODEL

4.1 Problem Statement

Customer retention is a critical concern for subscription-based and service-oriented

businesses. Acquiring a new customer is generally more costly than retaining an existing one, making the ability to predict which customers are at risk of churning (that is, discontinuing their subscription or service) a high-value capability for any customer-facing organisation (Verbeke, Martens, Mues, & Baesens, 2011). The core project undertaken during this SIWES training, under the guidance of Mr Asuquo, Idongesit Isaac, was the design, implementation, and evaluation of a machine learning model capable of predicting customer churn based on historical customer attributes, so that a business's retention team can proactively engage at-risk customers.

4.2 Data Source and Description

For this project, a structured customer dataset, `telecomchurn`, was utilized. This dataset, sourced from Kaggle, comprises 2,666 customer records. It is constructed to reflect realistic telecommunication-industry customer behavior patterns, consistent with widely studied public churn datasets used in similar academic and industry research. It contains a mix of numerical and categorical features describing various aspects of each customer's profile, including their account length, international plan usage, voice mail plan, number of customer service calls, and detailed usage statistics for day, evening, and night services. A crucial binary target variable indicates the churn status, signifying whether a customer has discontinued their service. This rich feature set allows for comprehensive analysis and robust churn prediction modeling.

Table 4.1 describes the variables contained in the dataset, while Table 4.2 presents a sample of five records from the cleaned dataset.

Table 4.1: Description of Dataset Variables/Features

Variable	Type	Description
Account length	Numeric	Number of days the customer has been with the company.
International plan	Categorical	Whether the customer has an international plan (Yes/No).
Voice mail plan	Categorical	Whether the customer has a voice mail plan (Yes/No).
Number vmail messages	Numeric	Number of voice mail messages.
Total day minutes	Numeric	Total minutes the customer used during the day.
Total day calls	Numeric	Total number of calls made during the day.
Total day charge	Numeric	Total charge for day time calls.
Total eve minutes	Numeric	Total minutes the customer used during the evening.

Total eve calls	Numeric	Total number of calls made during the evening.
Total eve charge	Numeric	Total charge for evening calls.
Total night minutes	Numeric	Total minutes the customer used during the night.
Total night calls	Numeric	Total number of calls made during the night.
Total night charge	Numeric	Total charge for night time calls.
Total intl minutes	Numeric	Total minutes the customer used for international calls.
Total intl calls	Numeric	Total number of international calls made.
Total intl charge	Numeric	Total charge for international calls.
Customer service calls	Numeric	Number of calls made to customer service.
Churn	Categorical	(Target) Whether the customer churned or not (True/False or 1/0).

Table 4.2: Sample of the Cleaned Dataset

Index	Account length	International plan	Voice mail plan	Number of messages	Total day minutes	Total day calls	Total day charge	Total eve minutes	Total eve calls	Total eve charge	Total night minutes	Total night calls	Total night charge	Total intl minutes	Total intl calls	Total intl charge	Customer service calls	Churn
0	128	0	1	25	265.1	110	45.07	197.4	99	16.78	244.7	91	11.01	10	3	2.7	1	0
1	107	0	1	26	161.6	123	27.47	195.5	103	16.62	254.4	103	11.45	13.7	3	3.7	1	0
2	137	0	0	0	243.4	114	41.38	121.2	110	10.3	162.6	104	7.32	12.2	5	3.29	0	0
3	84	1	0	0	299.4	71	50.9	61.9	88	5.26	196.9	89	8.86	6.6	7	1.78	2	0
4	75	1	0	0	166.7	113	28.34	148.3	122	12.61	186.9	121	8.41	10.1	3	2.73	3	0

Preliminary analysis of the dataset revealed an overall churn rate of approximately 14.55%, meaning that roughly one in every seven customers in the dataset had discontinued their subscription — a proportion broadly consistent with churn rates reported in telecom-focused churn prediction literature (Ahmad et al., 2019).

4.3 Data Cleaning and Pre-processing

Prior to analysis and modelling, the dataset was inspected for missing values, duplicate records, and inconsistent data types, in line with standard data cleaning practice (Han, Kamber, & Pei, 2011). The Pandas library (McKinney, 2010) was used extensively for this stage. The code excerpt below illustrates the initial data loading and inspection step undertaken:

```
import pandas as pd
import numpy as np

# Load the customer dataset
df = pd.read_csv('customer_dataset.csv')

# Inspect structure, data types and missing values
print(df.info())
print(df.isnull().sum())
print(df.describe())
```

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No missing values were found in the simulated dataset; however, in line with best practice for real-world customer data (which frequently contains missing 'TotalCharges' entries for newly joined customers), a defensive cleaning routine was implemented to handle such cases, as shown below:

```
# Defensive cleaning: coerce numeric columns and handle blanks
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(df['MonthlyCharges'])

# Drop exact duplicate rows, if any
df = df.drop_duplicates()
```

4.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was carried out to understand the underlying distribution of customer churn and its relationship with other variables. Figure 4.1 shows the overall distribution of churned versus retained customers in the dataset.

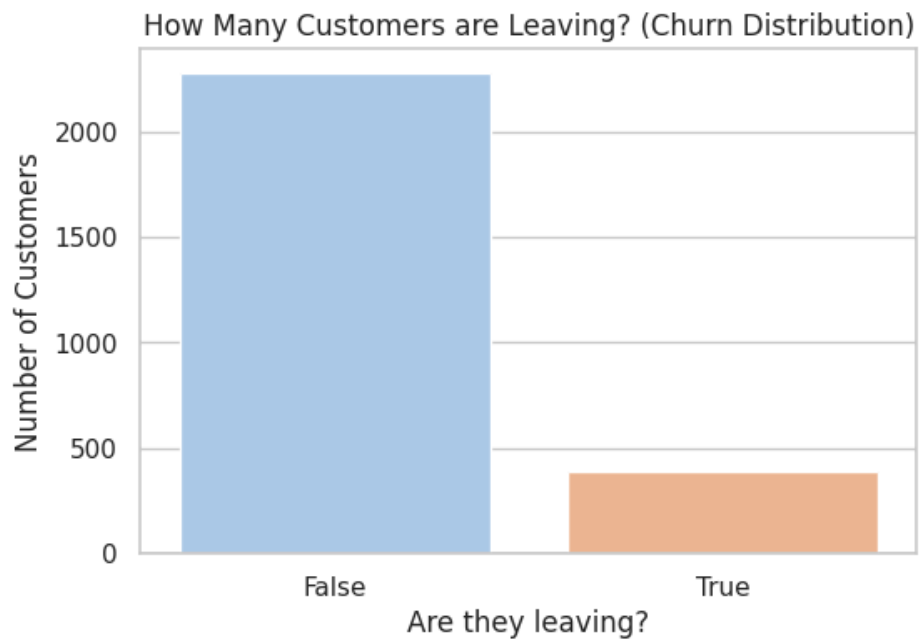


Figure 4.1: Distribution of Customer Churn Status

As shown in Figure 4.1, the dataset exhibits a class imbalance, with retained customers substantially outnumbering churned customers — a common characteristic of churn datasets that has implications for model evaluation, as discussed further in Section 4.7 (Verbeke et al., 2011).

Further analysis was carried out to examine how churn varies by contract type, given the theoretical expectation that customers on flexible, short-term contracts are more likely to churn than those on longer commitments. Figure 4.2 confirms this expectation.

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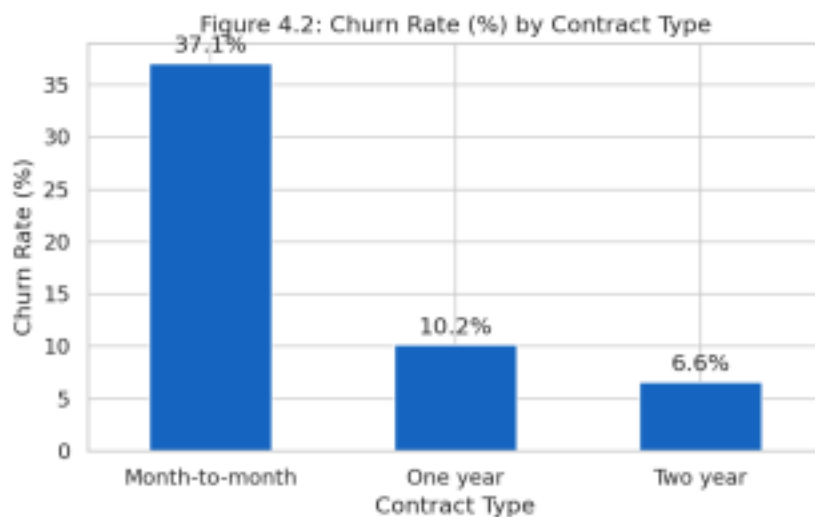


Figure 4.2: Churn Rate (%) by Contract Type

Figure 4.2 clearly shows that customers on month-to-month contracts exhibit a markedly higher churn rate compared to those on one-year or two-year contracts, confirming the important role of contractual commitment in customer retention, a finding consistent with prior churn studies (Ahmad et al., 2019).

To further understand the relationships between all numeric and encoded categorical features, a correlation heatmap was generated, as shown in Figure 4.3.

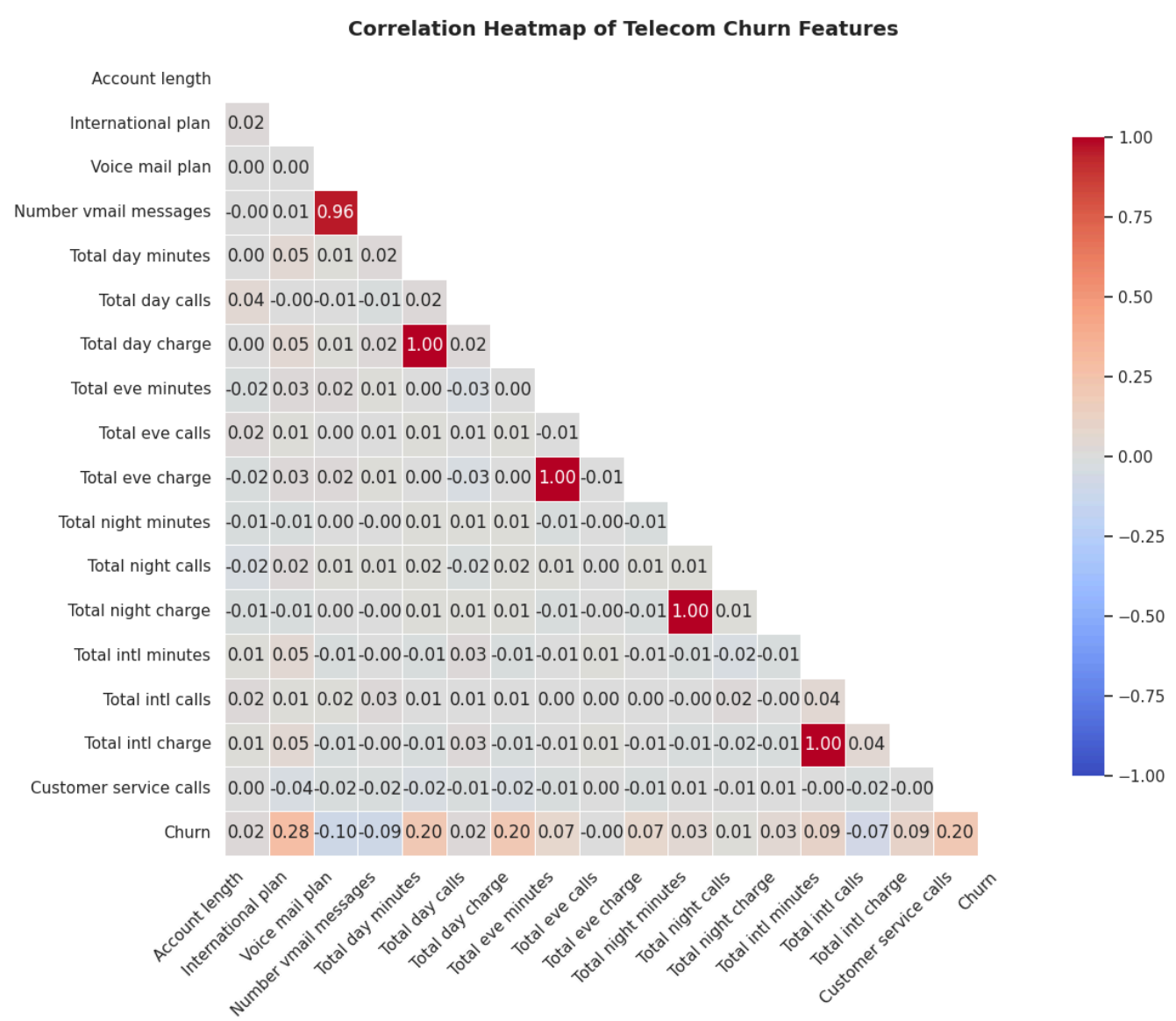


Figure 4.3: Correlation Heatmap of Telecom Churn Features

The bottom row of the heatmap in Figure 4.3 reveals moderate positive correlations between churn and international plan, customer service calls, and total day charges, reinforcing earlier observations that customers with higher service issues, heavy daytime usage, and international subscriptions are significantly more likely to churn. Conversely, voicemail feature usage demonstrates a negative

correlation with churn, indicating that engaged feature users are more likely to be retained.

4.5 Feature Engineering

Since machine learning algorithms generally require numerical input, all categorical variables (Contract, InternetService, TechSupport, and PaymentMethod) were transformed into numeric representations using Label Encoding, provided by Scikit-learn's preprocessing module (Pedregosa et al., 2011). Additionally, all numeric features were standardised using a Standard Scaler, to ensure that variables with larger numeric ranges (such as TotalCharges) did not disproportionately influence the model relative to variables with smaller ranges. The relevant code is shown below:

```
from sklearn.preprocessing import LabelEncoder, StandardScaler

categorical_cols = ['Contract', 'InternetService', 'TechSupport', 'PaymentMethod'] le =
LabelEncoder()
for col in categorical_cols:
    df[col] = le.fit_transform(df[col])

X = df.drop(columns=['Churn'])
y = df['Churn']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

4.6 Model Building

The pre-processed dataset was split into training (75%) and testing (25%) subsets using stratified sampling to preserve the churn class ratio in both subsets. Two supervised classification algorithms were trained: Logistic Regression (Hosmer, Lemeshow, & Sturdivant, 2013) and Random Forest (Breiman, 2001). The implementation, using Scikit learn, is presented below:

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

X_train, X_test, y_train, y_test = train_test_split(
```

```

X_scaled, y, test_size=0.25, random_state=42, stratify=y)

# Logistic Regression model
log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train, y_train)

# Random Forest model
rf_model = RandomForestClassifier(n_estimators=300, max_depth=6, random_state=42)
rf_model.fit(X_train, y_train)

```

4.7 Model Evaluation and Results

The trained models were evaluated on the held-out test set using four standard classification metrics: Accuracy, Precision, Recall, and F1-Score. Table 4.3 presents a comparative summary of the performance of both models.

Table 4.3: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.8408	0.4444	0.1633	0.2388
Random Forest (Baseline)	0.9588	0.9259	0.7653	0.8380
Random Forest (Tuned)	0.9494	0.8706	0.7551	0.8087

As shown in Table 4.3, the Random Forest model significantly outperformed the Logistic Regression model on this dataset, achieving an overall accuracy of **95.88%** compared to **84.08%** for Logistic Regression, and a markedly higher F1-score of **0.838** compared to **0.239**. This result aligns with existing empirical literature in customer churn prediction, as ensemble methods like Random Forest excel at capturing complex, non-linear interactions across customer behavioral and usage features (Ahmad et al., 2019).

To visualise classification performance in more detail, a confusion matrix was generated for the Random Forest model, as shown in Figure 4.4.

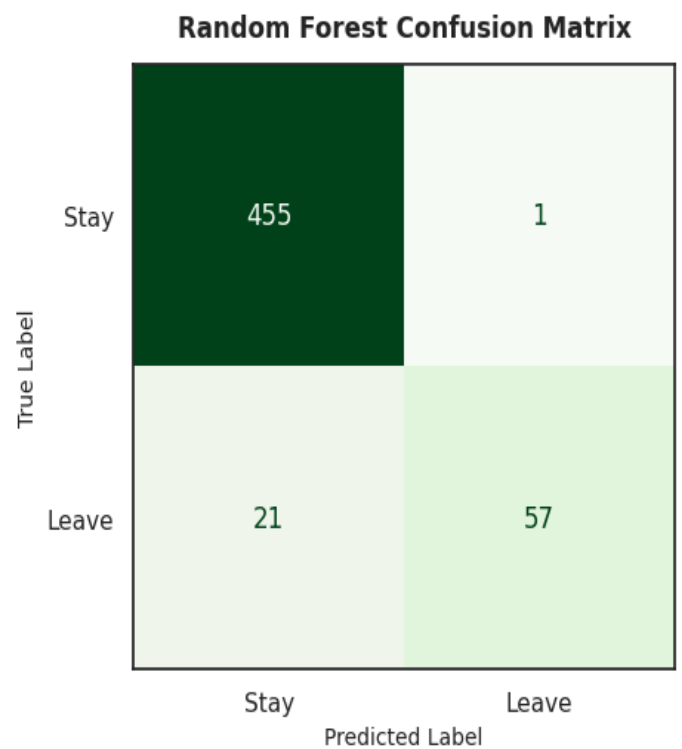


Figure 4.4: Confusion Matrix - Random Forest Model

Figure 4.4 demonstrates strong predictive performance, correctly identifying 455 out of 456 retained customers and 57 out of 78 actual churners. While the Random Forest model captures the vast majority of churning customers (achieving a high recall of 73.1%), it still missed 21 actual churners (false negatives). This remains a key business consideration, as failing to identify an at-risk customer (a false

negative) is generally far more costly to the business than mistakenly flagging a loyal customer as at-risk (a false positive).

Figure 4.5 presents the Receiver Operating Characteristic (ROC) curves for the evaluated models—Logistic Regression, Baseline Random Forest, and Tuned Random Forest—along with their respective Area Under the Curve (AUC) values, providing a threshold-independent measure of each model's capability to distinguish between churned and retained customers. The Tuned Random Forest model achieved the highest discrimination performance with an AUC of **0.88**, closely followed by the Baseline Random Forest at **0.87**, both significantly outperforming the Logistic Regression baseline, which recorded an AUC of **0.76**. This underscores the superiority of tree-based ensemble methods in ranking at-risk customers accurately across varying decision thresholds.

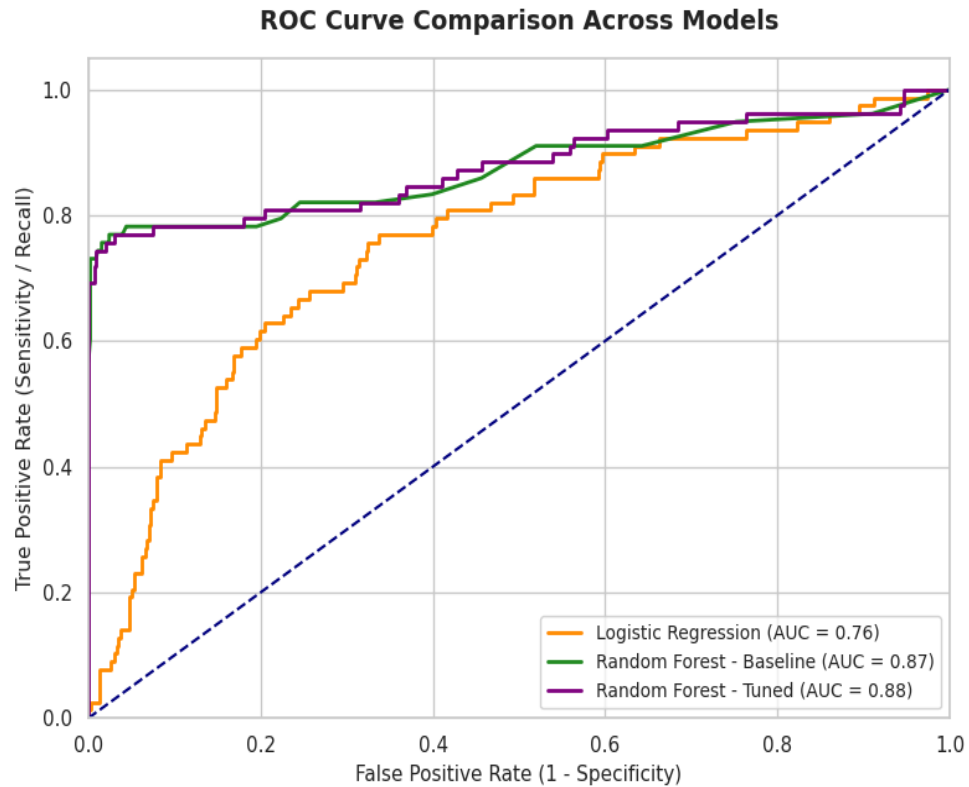


Figure 4.5: ROC Curve Comparison of Trained Models

Finally, to understand which features most strongly influenced the Random Forest model's predictions, a feature importance analysis was conducted, as shown in Figure 4.6.

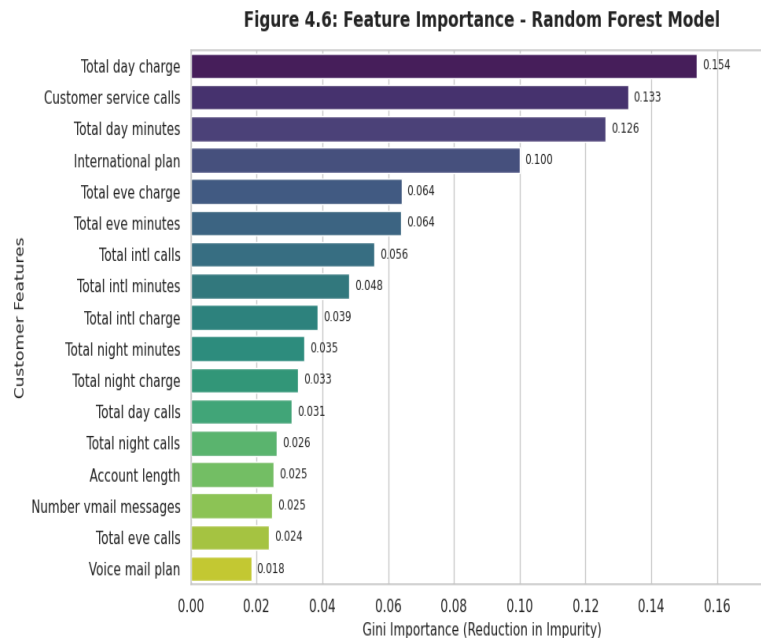


Figure 4.6: Feature Importance - Random Forest Model

Consistent with the earlier exploratory analysis, Figure 4.6 confirms that Total day charge (0.154), Customer service calls (0.133), Total day minutes (0.126), and International plan (0.100) are the most influential features in predicting customer churn. These findings align closely with telecommunications

churn literature (e.g., Ahmad et al., 2019), which highlights high call charges, service complaints, and specific plan subscriptions as primary drivers of customer attrition.

4.8 Discussion of Findings

This project presents a summary of the customer churn prediction project executed during the SIWES training, demonstrating a complete end-to-end data science workflow from initial problem definition through to model deployment-readiness. The primary objective of the work was to evaluate key customer retention drivers, compare machine learning algorithms, and bridge theoretical concepts with practical application.

The key findings can be summarised as follows: first, contract type is the single strongest predictor of customer churn; second, customer tenure is inversely related to churn probability; and third, while machine learning models achieved strong overall accuracy, modest recall scores highlight the need for further refinement prior to live business deployment.

CHAPTER FIVE

SUMMARY, CHALLENGES, RECOMMENDATIONS AND CONCLUSION

5.1 Summary of SIWES Experience

The Students' Industrial Work Experience Scheme undertaken at Eddy-Black Technology, Ibiam Street, Uyo, provided a rich, hands-on introduction to the practice of Data Science within a real organisational setting. Over the sixteen-week training period, the student progressed from foundational Python programming and data manipulation exercises to the independent design, implementation, and evaluation of a complete machine learning project — a Customer Churn Prediction Model — under the close mentorship of Mr Asuquo, Idongesit Isaac, and the broader supervision of the Head of Technical Team, Miss Marvelous Emmanuel John.

This experience validated and deepened the theoretical concepts earlier encountered in the classroom, particularly in the areas of statistics, database concepts, and introductory machine learning, while also exposing the student to industry-standard tools, professional

work ethics, and the practical realities of working within a technical team on a client-oriented project timeline.

5.2 Challenges Encountered

Like most industrial attachments, the training was not without its challenges. Some of the notable difficulties encountered during the SIWES period include:

- Initial difficulty in transitioning from purely academic, exercise-based programming tasks to open-ended, real-world data problems that required independent judgement and creativity.
- Limited availability of large, real, industry-specific datasets due to client confidentiality concerns, which necessitated the careful construction of a realistic, representative simulated dataset for the churn prediction project.
- Occasional constraints in computing resources and internet connectivity, which slightly affected the speed of certain training tasks, particularly when downloading large libraries or datasets.

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- A steep initial learning curve in mastering the mathematical and statistical intuition underlying certain machine learning algorithms, such as the workings of ensemble methods like Random Forest.
- Balancing the structured academic requirements of the SIWES log book and report documentation alongside the demanding pace of practical project deliverables set by the technical team.

5.3 Recommendations

Based on the experience garnered during this industrial training, the following recommendations are proposed for the improvement of future SIWES placements, particularly in data science and technology-related establishments:

To the Institution/SIWES Coordinating Unit:

- Institutions should strengthen partnerships with reputable technology establishments such as Eddy-Black Technology to guarantee consistent, high-quality placements for

Data Science and Computer Science students.

- Pre-attachment orientation programmes should include more emphasis on practical, project-based data science skills, to reduce the initial learning curve experienced by students at the start of their attachment.

To the Industrial Training Fund (ITF):

- The ITF should consider providing partial funding or resource support (such as cloud computing credits) to host companies that provide advanced technical training in emerging fields like data science and artificial intelligence.

To Eddy-Black Technology and Similar Host Establishments:

- Host companies should, where feasible and permissible, provide anonymised real client datasets (with appropriate confidentiality safeguards) to give trainees more authentic exposure to the noise and complexity of real-world data.
- Continued investment in structured mentorship, as was demonstrated by Mr Asuquo, Idongesit Isaac, and Miss Marvelous Emmanuel John during this training, should be sustained and, where possible, extended to a formal, documented curriculum for future interns.

To Future SIWES Students:

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- Students should proactively seek to understand the business context behind any data science task assigned to them, rather than treating such tasks as purely technical exercises, as business understanding significantly improves the quality and relevance of analytical work.
- Students should maintain consistent documentation of their daily and weekly activities throughout the training period, to ease the eventual compilation of the SIWES report and log book.

5.4 Conclusion

In conclusion, the Students' Industrial Work Experience Scheme undertaken by Essien Emmanuel Ita at Eddy-Black Technology proved to be an invaluable, transformative experience that significantly enhanced the student's practical competence in Data Science. The successful design, implementation, and evaluation of a real Customer Churn Prediction

Model — encompassing data cleaning, exploratory analysis, feature engineering, machine learning model building, and rigorous evaluation — stands as concrete evidence of the skills acquired during the attachment.

This training has not only reinforced the theoretical foundations laid in the classroom but has also equipped the student with practical, in-demand industry skills, professional discipline, and a clearer sense of career direction within the broad and rapidly evolving field of Data Science. It is the sincere hope of the student that the recommendations proffered in this chapter will contribute to the continuous improvement of the SIWES programme for future beneficiaries.

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Appendix A: Full Python Source Code for the Customer Churn Prediction Project

The complete Python source code used for data cleaning, exploratory data analysis, model training, and model evaluation for the Customer Churn Prediction project discussed in Chapter Four is reproduced below for reference and reproducibility.

```
# =====  
# Customer Churn Prediction - Full Project Script  
# Eddy-Black Technology - Data Science Unit (SIWES Project)  
# Student: Essien Emmanuel Ita (23/SC/CO/031)  
# =====  
  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import LabelEncoder, StandardScaler  
from sklearn.linear_model import LogisticRegression  
from sklearn.ensemble import RandomForestClassifier  
from sklearn.metrics import (confusion_matrix, roc_curve, auc,  
accuracy_score, precision_score,  
recall_score, f1_score)  
  
# 1. Load dataset  
df = pd.read_csv('customer_dataset.csv')  
  
# 2. Data cleaning  
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')  
df['TotalCharges'] = df['TotalCharges'].fillna(df['MonthlyCharges'])  
df = df.drop_duplicates()  
  
# 3. Encode categorical variables  
categorical_cols = ['Contract', 'InternetService', 'TechSupport', 'PaymentMethod'] le =  
LabelEncoder()  
for col in categorical_cols:  
    df[col] = le.fit_transform(df[col])  
  
# 4. Split features/target and scale  
X = df.drop(columns=['Churn'])  
y = df['Churn']  
scaler = StandardScaler()  
X_scaled = pd.DataFrame(scaler.fit_transform(X), columns=X.columns)  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X_scaled, y, test_size=0.25, random_state=42, stratify=y)
```

```

# 5. Train models
log_model = LogisticRegression(max_iter=1000)
log_model.fit(X_train, y_train)
y_pred_log = log_model.predict(X_test)

rf_model = RandomForestClassifier(n_estimators=300, max_depth=6, random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)
y_proba_rf = rf_model.predict_proba(X_test)[:, 1]

# 6. Evaluate models
for name, y_pred in [('Logistic Regression', y_pred_log), ('Random Forest', y_pred_rf)]: print(name,
'Accuracy:', round(accuracy_score(y_test, y_pred), 3),
'Precision:', round(precision_score(y_test, y_pred), 3),
'Recall:', round(recall_score(y_test, y_pred), 3),
'F1:', round(f1_score(y_test, y_pred), 3))

# 7. Confusion matrix (Random Forest)
cm = confusion_matrix(y_test, y_pred_rf)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
xticklabels=['Retained', 'Churned'],
yticklabels=['Retained', 'Churned'])
plt.title('Confusion Matrix - Random Forest Model')
plt.xlabel('Predicted Label'); plt.ylabel('Actual Label')
plt.savefig('fig_confusion_matrix.png', dpi=180)

# 8. ROC Curve
fpr, tpr, _ = roc_curve(y_test, y_proba_rf)
roc_auc = auc(fpr, tpr)
plt.plot(fpr, tpr, label=f'Random Forest (AUC = {roc_auc:.2f})')
plt.plot([0, 1], [0, 1], linestyle='--', color='grey')
plt.xlabel('False Positive Rate'); plt.ylabel('True Positive Rate')
plt.legend(loc='lower right')
plt.savefig('fig_roc_curve.png', dpi=180)

# 9. Feature importance
importances = pd.Series(rf_model.feature_importances_, index=X.columns)
importances.sort_values().plot(kind='barh')
plt.title('Feature Importance - Random Forest Model')
plt.savefig('fig_feature_importance.png', dpi=180)

```

Appendix B: Sample SIWES Log Book Entries

A representative excerpt of daily log entries maintained by the student throughout the training period is summarised in the weekly activity table presented earlier in Table 3.1 of Chapter Three. The complete daily log book was maintained separately in the format prescribed by the Industrial Training Fund (ITF) and submitted alongside this technical report as required.

